

# Solution Aggregations to Fall 2025 ICPC

## 1 NAEAST25

### 1.1 NAEAST25J

Given a string  $s$ , find:

1. ways to write  $s$  as a cyclic shift of  $xy\overline{xy}$ .
2. ways to write  $s$  as a cyclic shift of  $xyz\overline{xyz}$ .

Here  $x$ ,  $y$ , and  $z$  are strings, and  $\overline{s}$  means the reversal of  $s$ .

#### 1.1.1 $O(n^2 \log n)$ solution

We solve in  $O(n \log n)$  for a fixed cyclic rotation.

A problem that uses the exact same techniques: CF906E, <https://codeforces.com/contest/906/problem/E>.

Let the characters of  $s$  be

$$s \triangleq s_1 s_2 \dots s_{2n},$$

consider the string  $t$  formed by interleaving the first and second halves of  $s$ :

$$t = s_1 s_{n+1} s_2 s_{n+2} \dots s_n s_{2n}.$$

This has the advantage that matches in the two halves is equivalent a part of  $t$  being a palindrome.

**Lemma 1.1.** *For any substring of the first  $n$  characters,  $1 \leq x \leq y \leq n$ ,  $s[x \dots y] = \overline{s[x + n \dots y + n]}$  if and only if*

$$t[2x - 1 \dots 2y]$$

*is a palindrome.*

*Proof.* The condition needed is for each  $0 \leq i < y - x$ ,

$$s[x + i] = s[y + n - i]$$

but by construction of  $t$ , we have

$$s[x + i] = t[2x - 1 + 2i] \tag{1}$$

$$s[y + n - i] = t[2y - 2i] \tag{2}$$

so this is equivalent to

$$t[2x - 1 + 2i] = t[2y - 2i]$$

for all  $i$ . We get this for all  $j$  because the odd even index ones can be obtained by going backwards on  $t$ .  $\square$

So, the problem is equivalent to counting the ways to partition  $t$  into 2 or 3 even palindromes. This is a classical topic, see e.g.

1. <https://acm.timus.ru/problem.aspx?space=1&num=1954&locale=en> from 2013 for cutting into 5 palindromes.
2. <https://acm.timus.ru/contest.aspx?id=276> from 2015 for cutting into arbitrary amounts of palindromes plus more stuff

The main idea is that we can perform the following DP transition on palindromes only:

**Lemma 1.2.** *Given string  $s[1 \dots n]$  and array  $f[1 \dots n]$ , we can compute in  $O(n \log n)$  time  $g[1 \dots n]$  where*

$$g[i] = \sum_{j \leq i, s[j \leq i] = \overline{s[j \leq i]}} f[j].$$

We can use this by:

1. obtain the ‘even length’ condition by setting all the odd indexed characters to a special character.
2. Run this two times for  $xy\overline{xy}$ , three times for  $xyz\overline{xyz}$ .

*Proof.* (of Lemma 1.2) **Richard:** [this is a ‘standard black box’, so I didn’t edit it too heavily...](#)

By the Palindrome Automaton (PAM) one can decompose all palindrome suffixes of a prefix of  $t$  into  $O(\log n)$  arithmetic sequences:

1. if  $s[p, i], s[p + d, i], \dots, s[p + kd, i]$  are all palindromes and respectively correspond to vertices  $x_0, x_1, \dots, x_k$  on the PAM,
2. one can deduce that

$$s[p + jd, i - d] = s[p + (j + 1)d, i]$$

for all  $0 \leq j < k$ ,

3. so only (the deepest) vertex  $x_0$  is newly introduced by position  $i$ .

We record  $\sum_{q=0}^j f[p + qd]$  on the vertex  $x_{k-j}$  for all  $0 \leq j \leq k$ : Note that the order we maintain this sum is the reverse order of the palindromes that the vertices really represent).

1. At position  $i$  we only need to update  $V(x_0)$ , the value recorded on the newly added vertex, as  $V(x_0) = V(x_1) + f[p + kd]$ . This is because  $V(x_1), \dots, V(x_k)$  are already correctly maintained at position  $i - d$ .
2. Then we just add  $g[i]$  by  $V(x_0)$  for all the  $x_0$ ’s that correspond to the longest palindrome in such an arithmetic sequence.

$\square$